LECTURE #5
LATERAL DYNAMICS

Automotive Engineering

KNUT

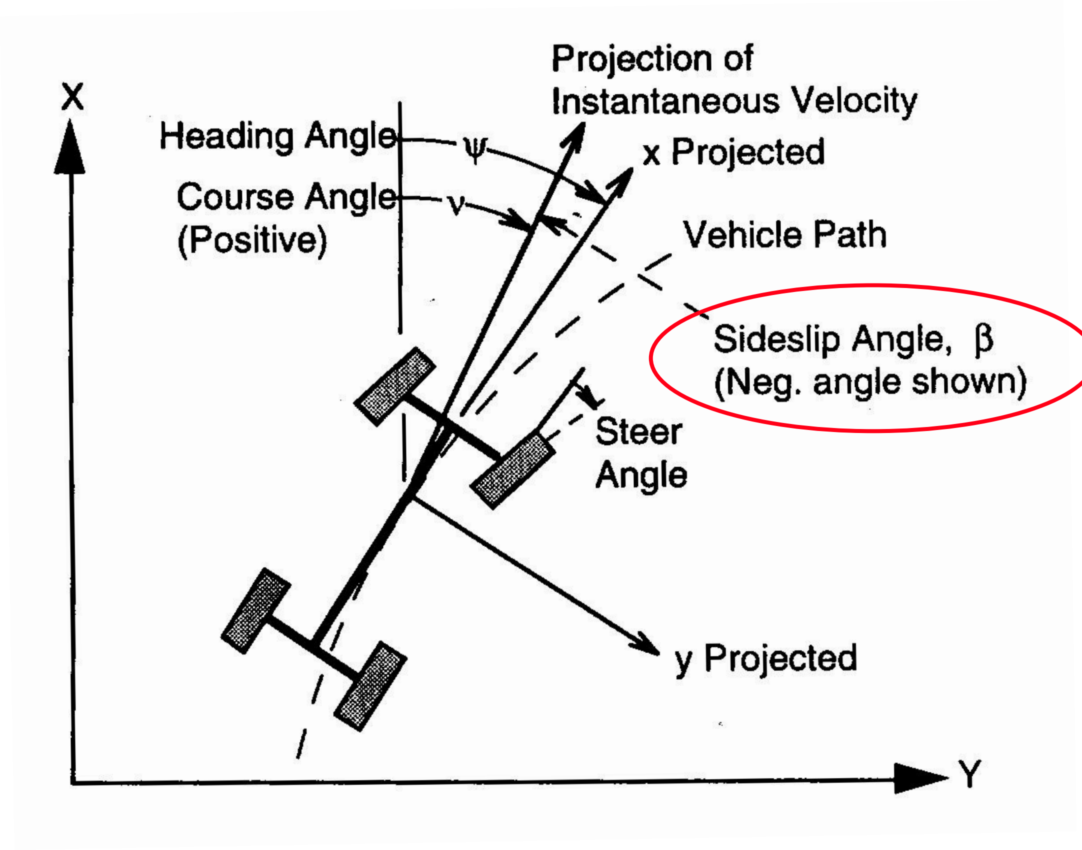
Spring, 2024

Lateral Dynamics

- Trivia
  - Lane departure: More than 39% of crash-related fatalities
  - Yaw dynamics is under damped
    - Less damping at a higher speed
  - Yaw dynamics control (e.g. ESP) helps
  - But, physics rules
  - Four wheel steering helps, but at an extra cost

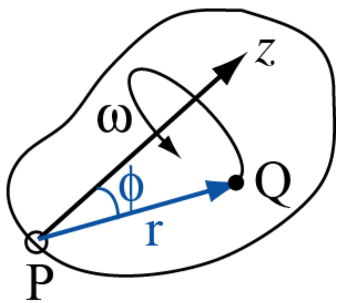
Lateral Dynamics

- Earth fixed coordinate system



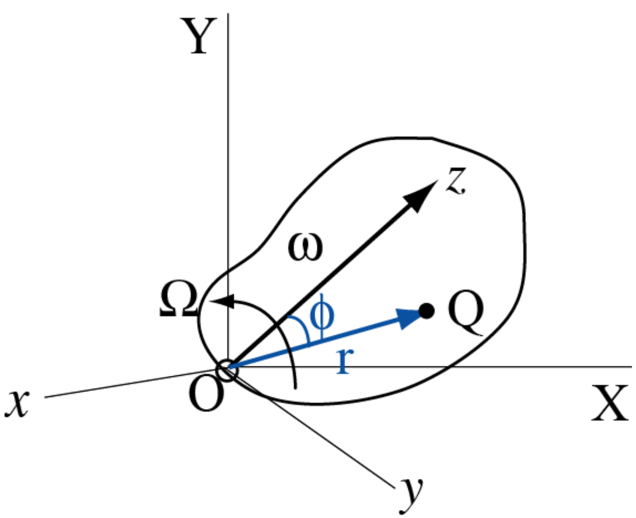
Lateral Dynamics

- Motion of a **static particle Q** in a rotating frame(xyz), at fixed frame(XYZ)



$$v_Q = \frac{d\mathbf{r}}{dt} = \frac{ds}{dt}\cdot\mathbf{i}_{\text{tangent to path Q}} = r\sin\phi\,\omega\cdot\mathbf{i} = \boldsymbol{\omega}\times\mathbf{r}$$

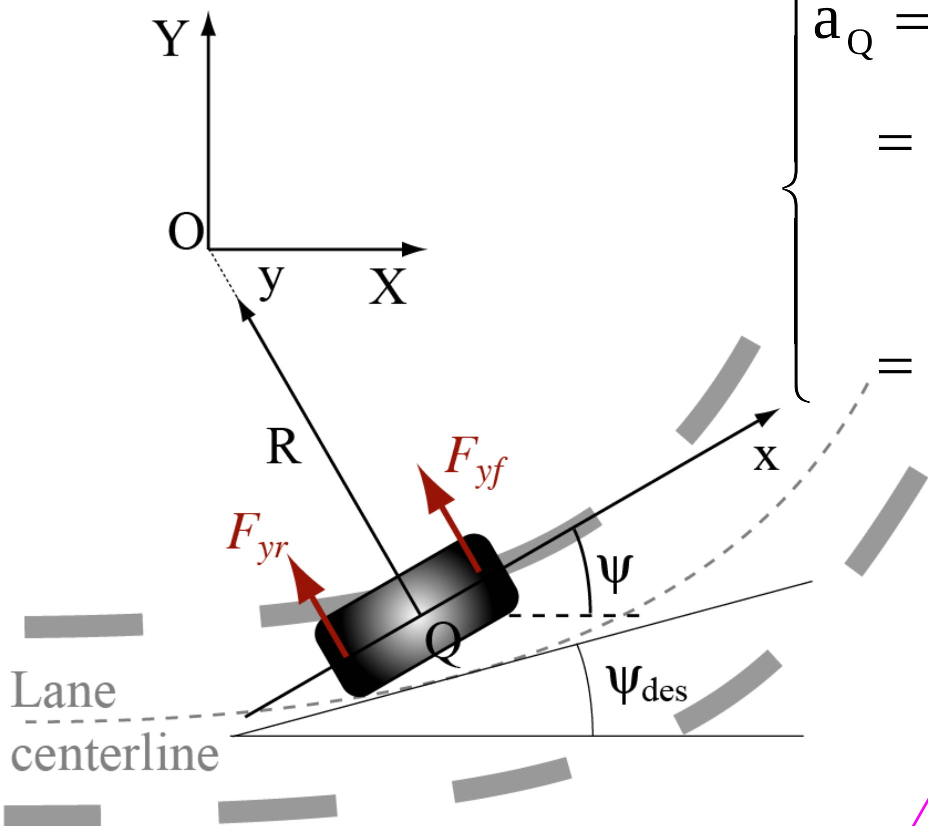
- Motion of a **moving particle Q** in a rotating frame, at fixed frame



$$\begin{cases} v_Q = \dot{Q}_{XYZ} = \Omega\times r + \dot{r}_{xyz} \quad \left(\left.\dfrac{d\mathbf{r}}{dt}\right|_{XYZ} = \left.\dfrac{dr_{xyz}}{dt}\right|_{XYZ} + \left.\dfrac{dr_{xyz}}{dt}\right|_{xyz} \right) \\ a_Q = \Omega\times\left(\Omega\times r + \dot{r}_{xyz}\right) + \left.\dfrac{d}{dt}\left(\Omega\times r + \dot{r}_{xyz}\right)\right|_{xyz} \\ \quad = \Omega\times\left(\Omega\times r\right) + \Omega\times\dot{r}_{xyz} + \dot{\Omega}\times r + \Omega\times\dot{r}_{xyz} + \ddot{r}_{xyz} \\ \quad = \dot{\Omega}\times r + \Omega\times\left(\Omega\times r\right) + 2\Omega\times\dot{r}_{xyz} + \ddot{r}_{xyz} \end{cases}$$

Lateral Dynamics

- Planar motion of vehicle C.G. about a fixed frame



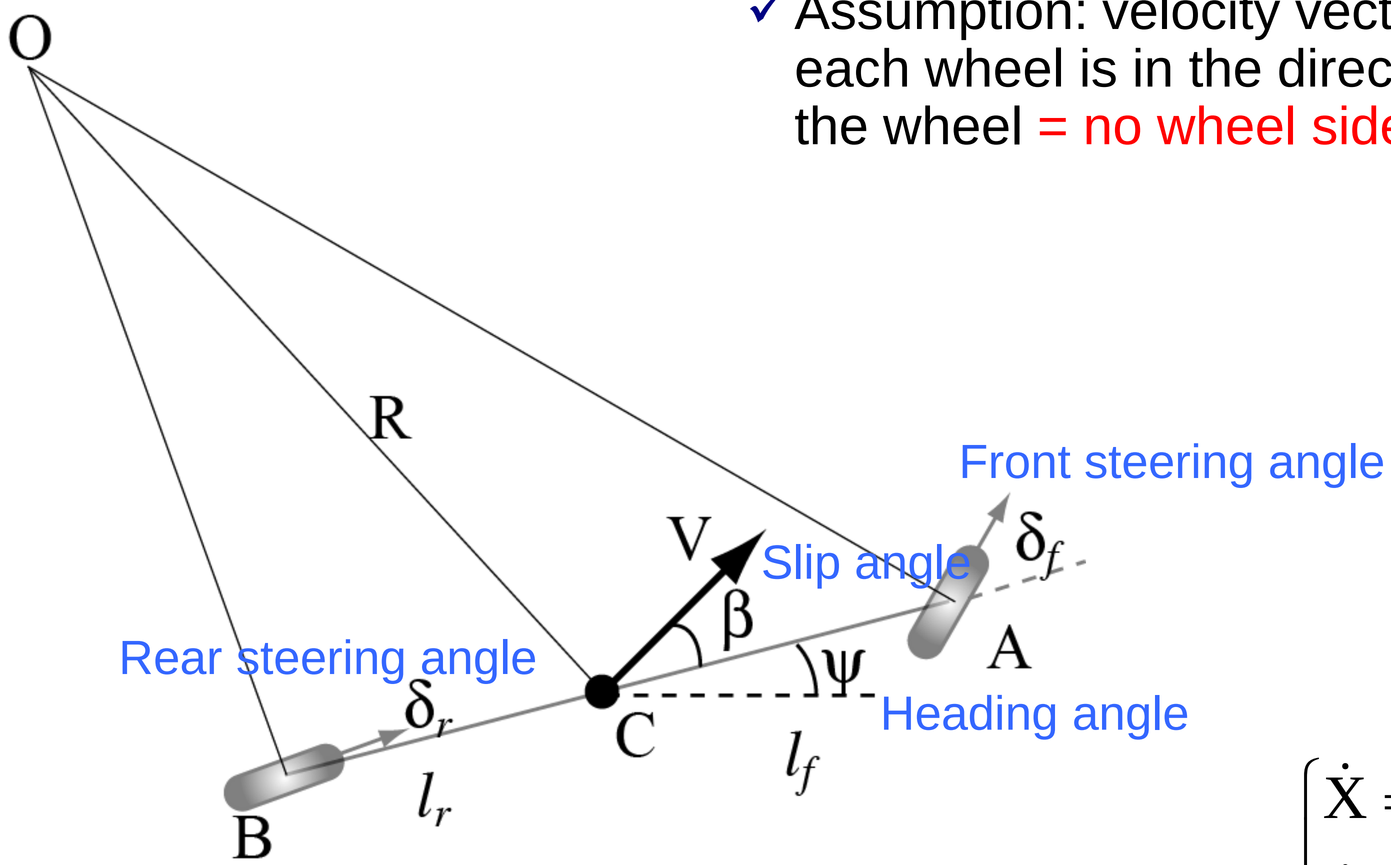
$$\begin{cases} a_Q = \dot{\Omega}\times r + \Omega\times(\Omega\times r) + 2\Omega\times\dot{r}_{xyz} + \ddot{r}_{xyz} \\ \quad = \ddot{\psi}\hat{k}\times(-R\hat{j}) + \dot{\psi}\hat{k}\times(\dot{\psi}\hat{k}\times(-R\hat{j})) + \cdots \\ \quad\quad 2\dot{\psi}\hat{k}\times(-\dot{R}\hat{j}) + (\ddot{x}\hat{i} + \ddot{y}\hat{j}) \\ \quad = (\ddot{\psi}R + 2\dot{\psi}\dot{R} + \ddot{x})\hat{i} + (\dot{\psi}^2 R + \ddot{y})\hat{j} \end{cases}$$

$$\Rightarrow \quad a_y = (\dot{\psi}^2 R + \ddot{y}) = \dot{\psi}\cdot\dot{\psi}R + \ddot{y} = \dot{\psi}\cdot V_x + \ddot{y}$$

Lateral Kinematics Model

- Equation of motion

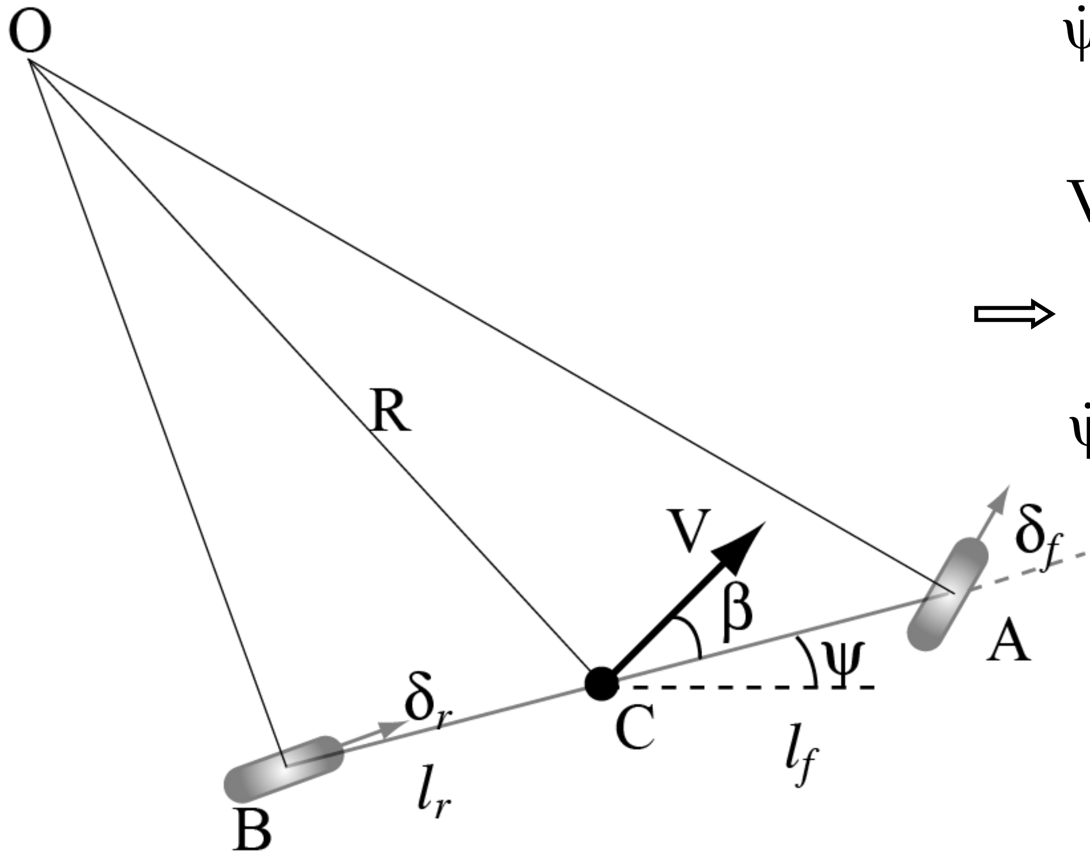
✓ Assumption: velocity vector at each wheel is in the direction of the wheel = no wheel side slip



$$\begin{cases} \dot{X} = V\cos(\psi + \beta) \\ \dot{Y} = V\sin(\psi + \beta) \\ \dot{\psi} = ? \end{cases}$$

Lateral Kinematics Model

- Equation of motion



$$\dot{\psi} = \frac{V_A \sin\delta_f - V_B \sin\delta_r}{l_f + l_r}$$

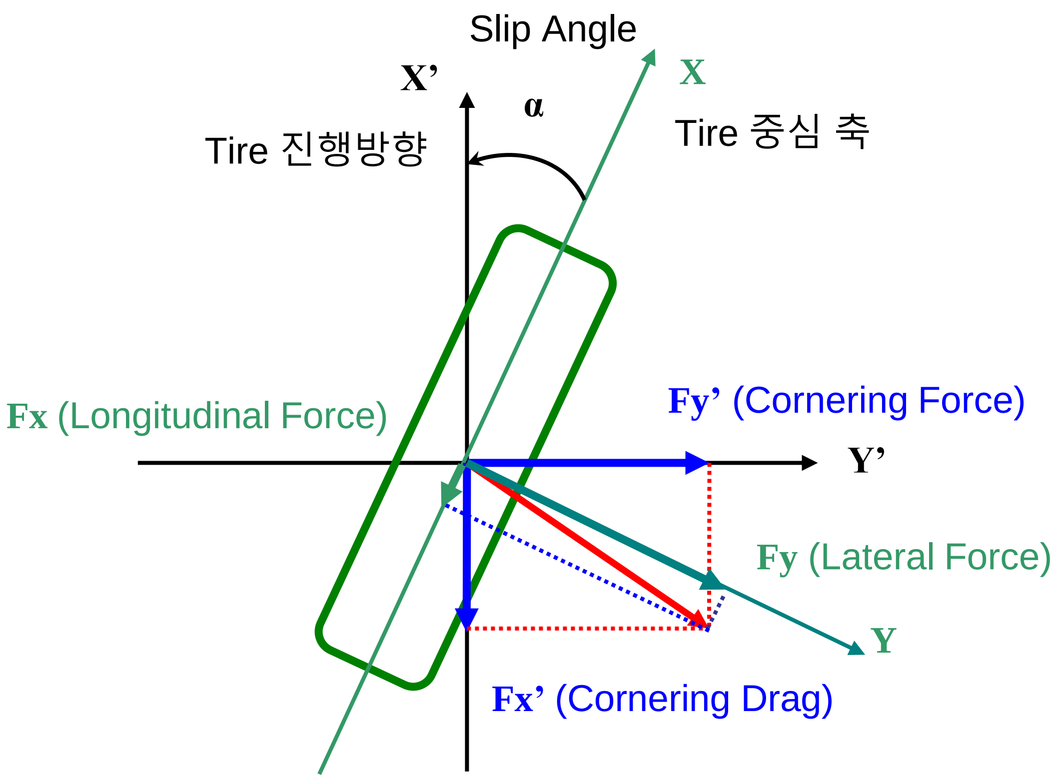
$$V_A \cos\delta_f = V_B \cos\delta_r = V\cos\beta$$

$$\Longrightarrow$$

$$\dot{\psi} = \frac{V\cos\beta}{l_f + l_r}\left\{\tan\delta_f - \tan\delta_r\right\}$$

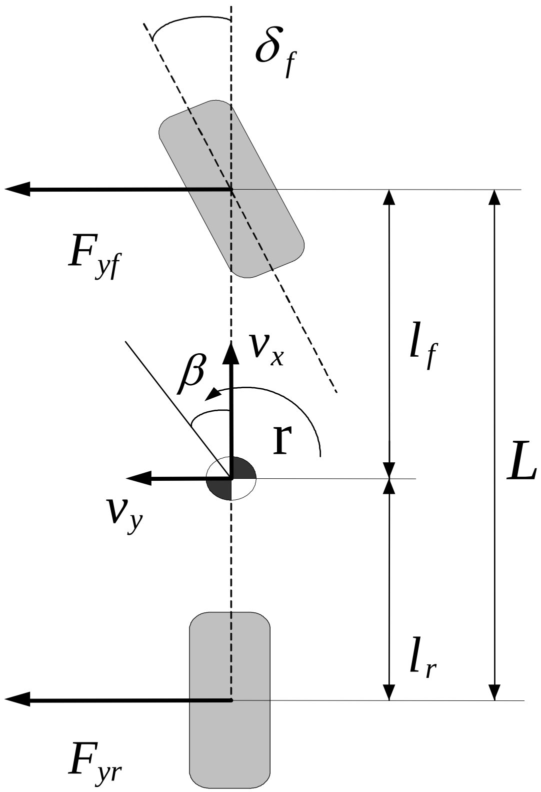
Lateral Dynamics Model

- Definition of tire force



Lateral Dynamics Model

- Dynamic model in terms of yaw rate and body side slip angle: for front wheel steering vehicle & flat surface
  - Neglect body roll dynamics
  - Neglect side-to-side weight shifting
  - Assume wheels are located at the vehicle center line



$$\text{Note: } r = \dot{\psi}$$

✓Called linear bicycle model (or single-track model)

Lateral Dynamics Model

- Dynamic model in terms of yaw rate and body side slip angle: for front wheel steering vehicle & flat surface

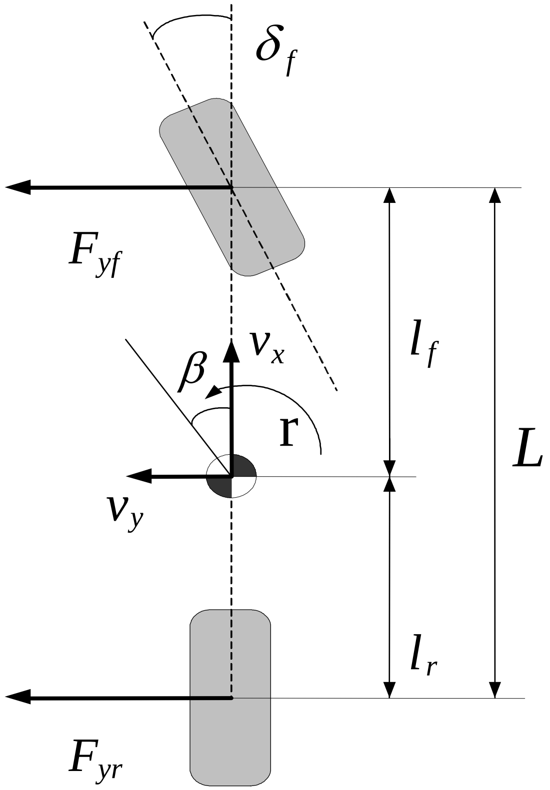
- Body sideslip angle

$$\beta \equiv \frac{v_y}{v_x}$$

- Wheel sideslip angle

$$\alpha_r \equiv \frac{v_{yr}}{v_{xr}} = \frac{v_y - l_r r}{v_x} = \beta - \frac{l_r}{v_x} r$$

$$\alpha_f \equiv \frac{v_{yf}}{v_{xf}} = \frac{v_y + l_f r - v_x \delta_f}{v_x} = \beta + \frac{l_f}{v_x} r - \delta_f$$



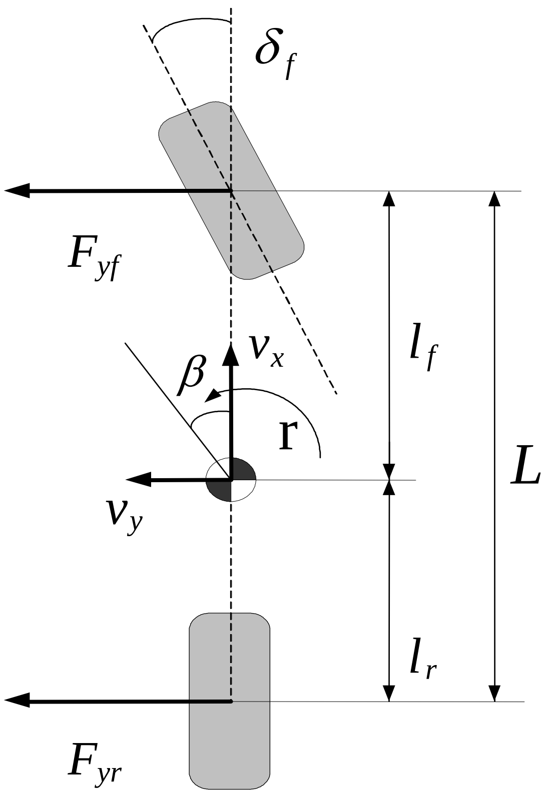
Lateral Dynamics Model

- Dynamic model in terms of yaw rate and body side slip angle: for front wheel steering vehicle & flat surface

Linear tire model

$$F_{yf} \equiv -C_f\,\alpha_f = -C_f\left(\beta + \frac{l_f}{v_x} r - \delta_f\right)$$

$$F_{yr} \equiv -C_r\,\alpha_r = -C_r\left(\beta - \frac{l_r}{v_x} r\right)$$



✓ $\mathbf{F_y}$ and $\boldsymbol{\alpha}$ have opposite signs

✓ $\mathbf{C_{f/r}}$: cornering stiffness of each axle = two times of the cornering stiffness of each wheel, function of tire normal load

Lateral Dynamics Model

- Dynamic model in terms of yaw rate and body side slip angle: for front wheel steering vehicle & flat surface
- Force balance:

$$m(\ddot{y} + r v_x) = F_{yf} + F_{yr}$$

- Moment balance:

$$I_z \dot{r} = F_{yf} l_f - F_{yr} l_r$$

- For constant or slowly varying vehicle speed

$$\Rightarrow \quad \dot{\beta} = \ddot{y}/v_x$$

$$\dot{\beta} = -r + \frac{C_f}{mv_x}\left(-\beta - \frac{l_f}{v_x} r + \delta_f\right) + \frac{C_r}{mv_x}\left(-\beta + \frac{l_r}{v_x} r\right)$$

$$\dot{r} = \frac{C_f l_f}{I_z}\left(-\beta - \frac{l_f}{v_x} r + \delta_f\right) - \frac{C_r l_r}{I_z}\left(-\beta + \frac{l_r}{v_x} r\right)$$

Lateral Dynamics Model

- Dynamic model in terms of yaw rate and body side slip angle: for front wheel steering vehicle & flat surface
- Or

$$\dot{\beta} = -\frac{C_f + C_r}{m v_x}\beta + \left(\frac{C_r l_r - C_f l_f}{m v_x^2} - 1\right) r + \frac{C_f}{m v_x}\delta_f$$

$$\dot{r} = \frac{C_r l_r - C_f l_f}{I_z}\beta - \frac{C_f l_f^2 + C_r l_r^2}{I_z v_x} r + \frac{C_f l_f}{I_z}\delta_f$$

✓ Parameters are constants except $\mathbf{v_x}$

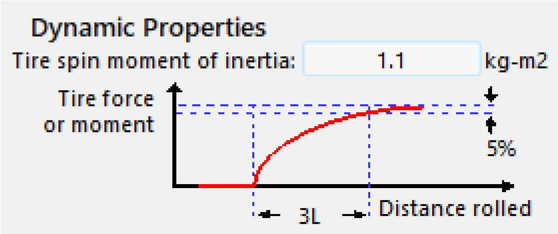
✓ Lateral dynamics is velocity dependent

Lateral Dynamics Model

- Dynamic tire model

$$\tau \dot{F}_{y\_lag} + F_{y\_lag} = F_y$$

$$\tau = \frac{C_\alpha}{K V_x}, \; L = \frac{C_\alpha}{K} \approx 1.5\text{m}, \; K = \left.\frac{\partial F_y}{\partial y}\right|_{y=0}$$



L = Relaxation length, K = Lateral stiffness

- Laplace transform

$$\tau s F_{yf\_lag} + F_{yf\_lag} = F_{yf}$$

$$F_{yf\_lag} = \frac{C_f\left(\delta_f - \left(\beta + \frac{l_f r}{v_x}\right)\right)}{\tau s + 1}$$

$$\tau s F_{yr\_lag} + F_{yr\_lag} = F_{yr}$$

$$F_{yr\_lag} = \frac{C_r\left(-\beta + \frac{l_r r}{v_x}\right)}{\tau s + 1}$$

Lateral Dynamics Model

- Bicycle model with dynamic tire model

$$mv_x(s\beta+r)=F_{yf\_lag}+F_{yr\_lag}$$

$$I_z s r = l_f F_{yf\_lag} - l_r F_{yr\_lag}$$

$$\Longrightarrow$$

$$mv_x(s\beta+r)(\tau s+1) = mv_x\left(s^2\tau\beta+(\beta+r\tau)s+r\right)$$

$$= C_f\left(\delta_f - \left(\beta+\frac{l_f r}{v_x}\right)\right)+C_r\left(-\beta+\frac{l_r r}{v_x}\right)$$

$$I_z sr(\tau s+1) = I_z\left(\tau s^2 r+sr\right)$$

$$= l_f C_f\left(\delta_f - \left(\beta+\frac{l_f r}{v_x}\right)\right)-l_r C_r\left(-\beta+\frac{l_r r}{v_x}\right)$$

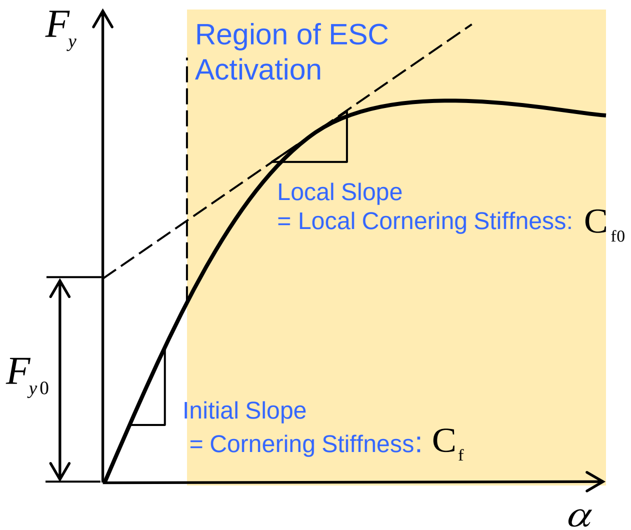
- In state space form

$$x = \begin{bmatrix}\beta & \dot{\beta} & r & \dot{r}\end{bmatrix}^T$$

$$\dot{x} = A\dot{x}+B\delta$$

Accounting for Nonlinearities

- Bicycle model with nonlinear tire model: locally linearized



$$F_{yf} = C_f \alpha_f$$

$$\Downarrow$$

$$F_{yf} = C_{f0}\alpha_f + F_{y0}$$

$$\dot{x} = A\dot{x} + B\delta + \boxed{E_{add}}$$

- In state space form

$$\dot{\beta} = -r + \frac{1}{mv_x}\left(C_{f0}\left(-\beta - \frac{l_f}{v_x}r + \delta_f\right) + F_{0f}\right) + \frac{1}{mv_x}\left(C_r\left(-\beta + \frac{l_r}{v_x}r\right) + F_{0r}\right)$$

$$\dot{r} = \frac{l_f}{I_z}\left(C_{f0}\left(-\beta - \frac{l_f}{v_x}r + \delta_f\right) + F_{0f}\right) - \frac{l_r}{I_z}\left(C_{r0}\left(-\beta + \frac{l_r}{v_x}r\right) + F_{0r}\right)$$

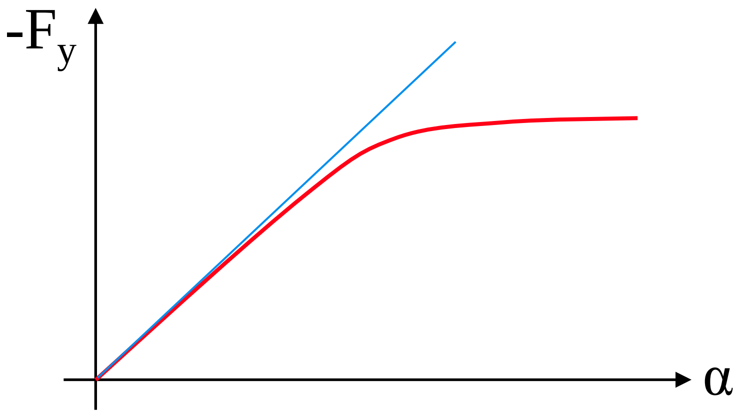
Lateral Dynamics

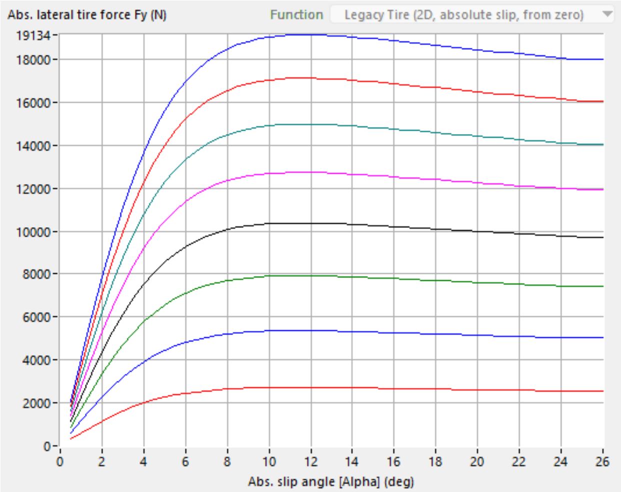
- Nonlinear effects
- Tire lateral force is NOT a linear function of tire side slip angle
- Tire lateral force is saturated at a large side slip angle

$$F_{yf} \neq -C_f\,\alpha_f\,,\; F_{yf} = C_f(\alpha_f)$$

$$F_{yr} \neq -C_r\,\alpha_r\,,\; F_{yr} = C_r(\alpha_r)$$

✓ Real vehicle dynamics can not be expressed as a linear explicit function of β & r



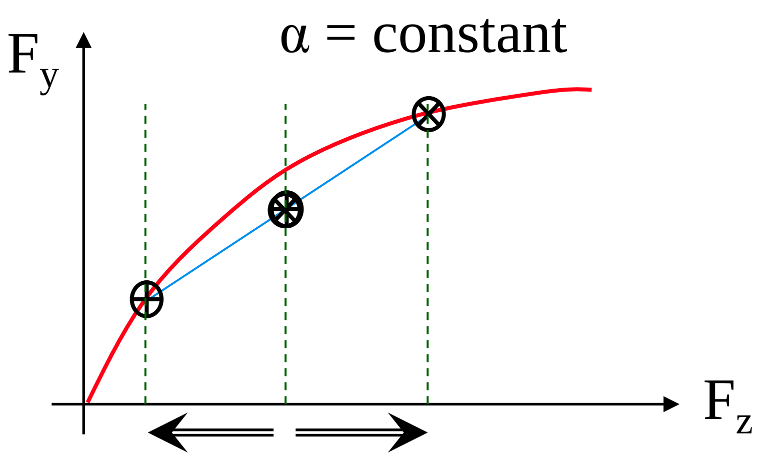


Lateral Dynamics

- Nonlinear effects
  - Tire lateral force is not proportional to the normal load
  - Tire-to-surface friction coefficient is reduced with load
- For small side slip angle: $F_{y\,f/r} = -C_{f/r}\,\alpha_{f/r}$

✓ $C_{f/r}$ is still a function of the normal load

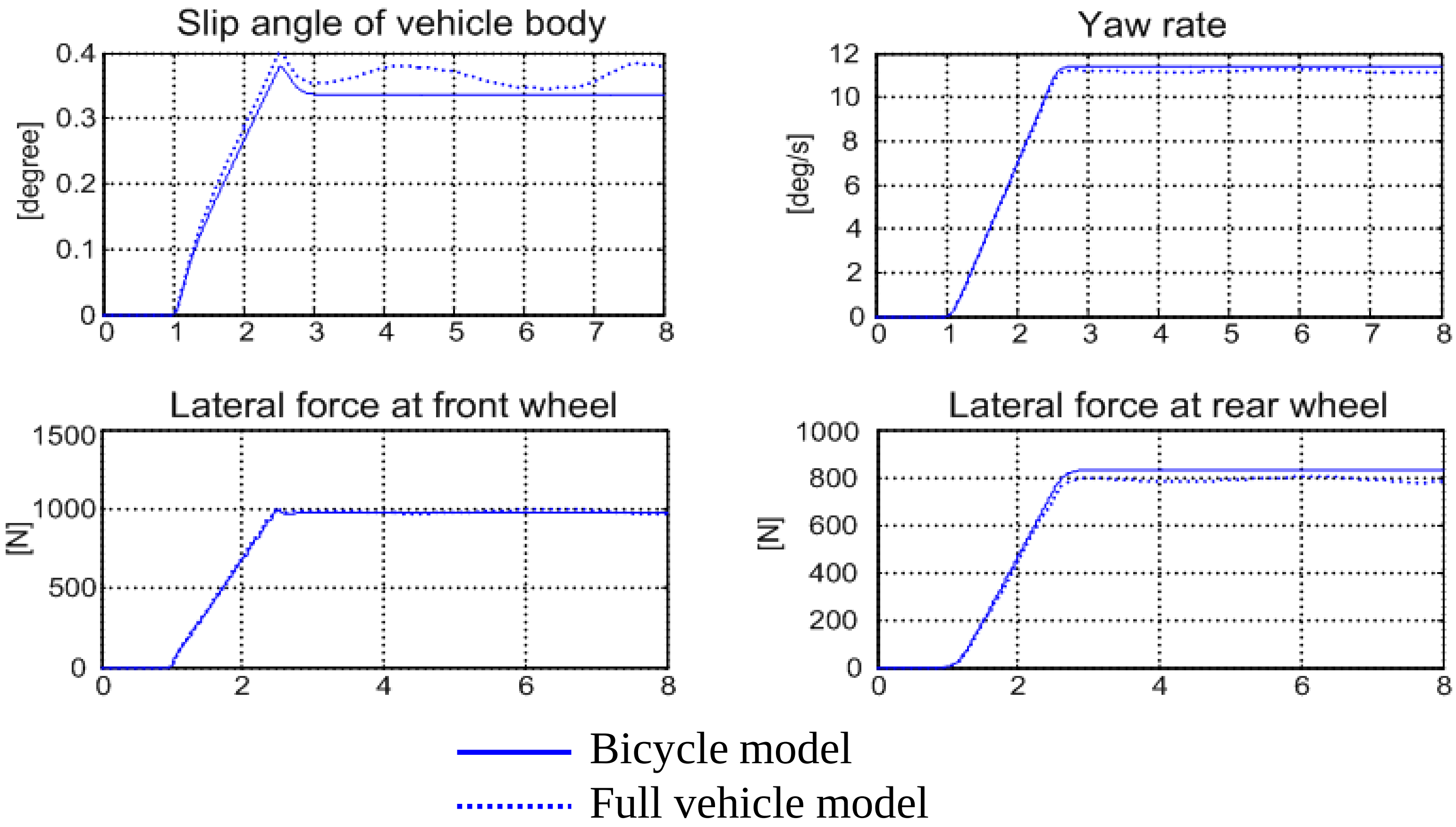
✓ More side-to-side weight shifting → reduced lateral friction coefficient



✓ Stiffer roll bar → more weight shifting

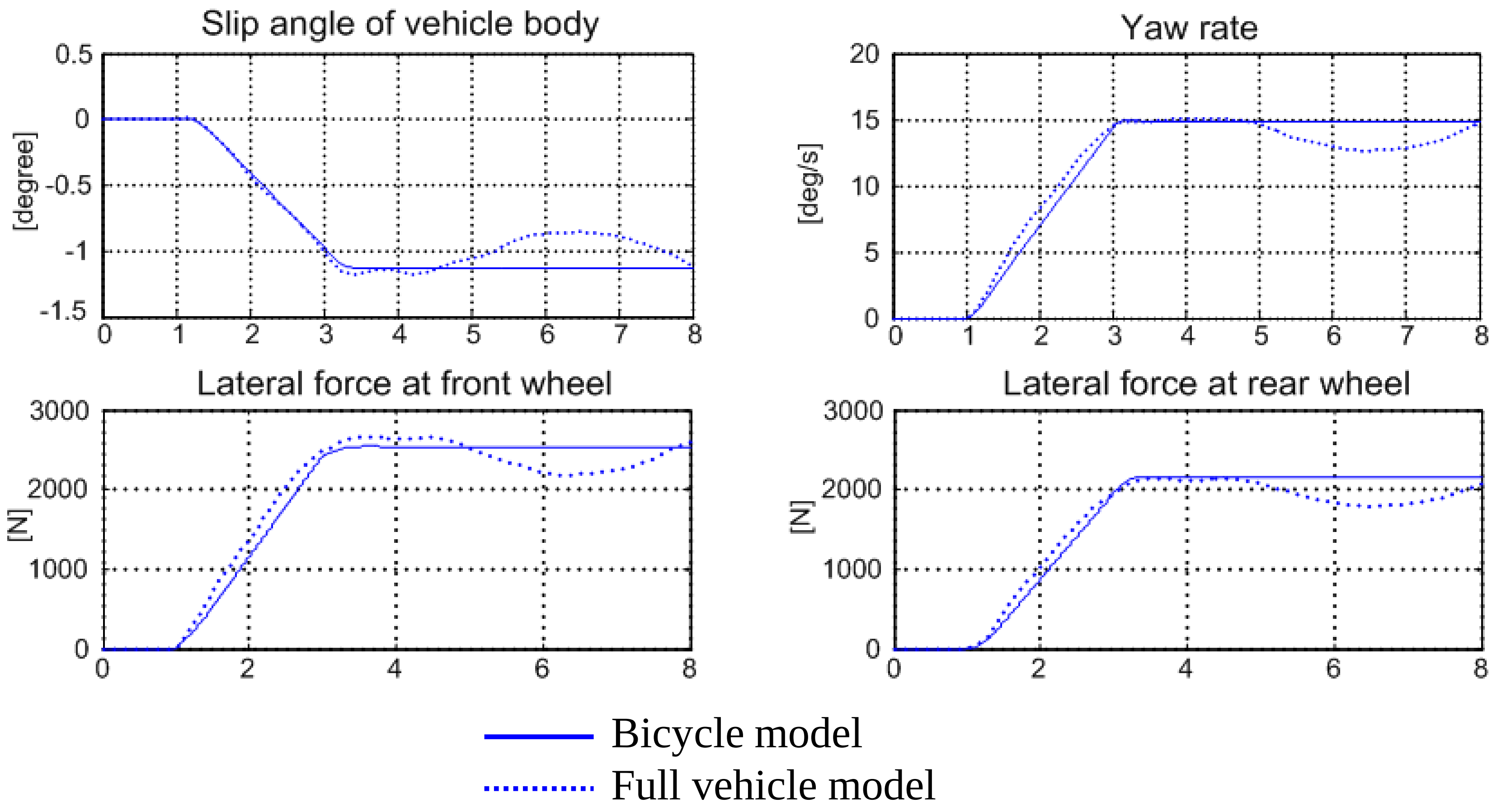
Bicycle Model vs. Full Vehicle (Nonlinear) Model

- Vehicle velocity: 40 km/h
- Steer angle: 3 degree



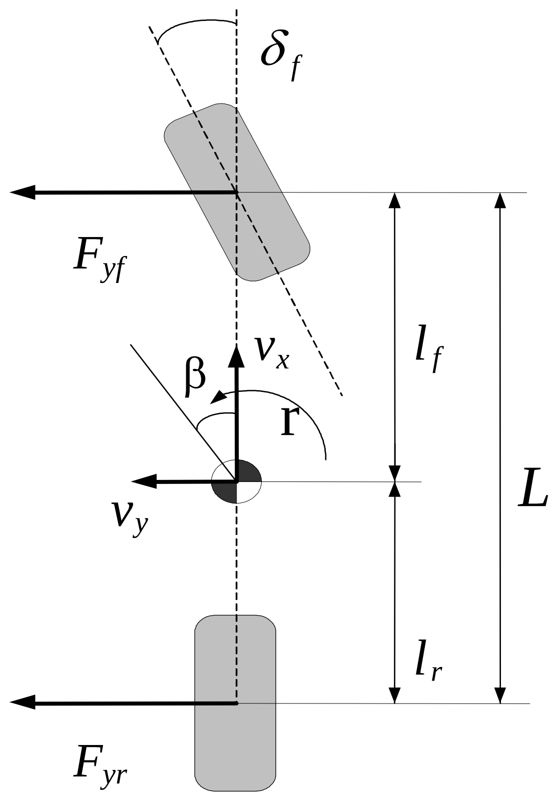
Bicycle Model vs. Full Vehicle (Nonlinear) Model

- Vehicle velocity: 80 km/h
- Steer angle: 3 degree



Understeer Gradient

- Steady-state cornering(Uniform circular motion):



$$\dot{\vec{v}} = 0 \ \left(\dot{v}_x = \dot{v}_y = 0\right) \rightarrow m\frac{v_x^2}{R} = F_{yf} + F_{yr}$$

$$\dot{r} = 0 \rightarrow 0 = l_f F_{yf} - l_r F_{yr}$$

$$F_{yf} = m\frac{l_r}{L}\frac{v_x^2}{R}, \quad F_{yr} = m\frac{l_f}{L}\frac{v_x^2}{R}$$

Also,

$$F_{yf} = -C_f \alpha_f , \quad F_{yr} = -C_r \alpha_r$$

Since,

$$\alpha_f = \beta + \frac{l_f}{v_x} r - \delta_f , \quad \alpha_r = \beta - \frac{l_r}{v_x} r$$

$$\Rightarrow \delta_f = \frac{L}{v_x} r - \alpha_f + \alpha_r = \frac{L}{R} - \alpha_f + \alpha_r$$

$$= \frac{L}{R} + \left(\frac{ml_r}{C_f L} - \frac{ml_f}{C_r L}\right)\frac{v_x^2}{R} \equiv \frac{L}{R} + K_v \frac{v_x^2}{R}$$

✓ K_v: Understeer gradient

Under / Neutral / Over-steer

$$K_v > 0 \quad \Leftrightarrow \quad |\alpha_f| > |\alpha_r| \quad \cdots \quad \text{Under-steer}$$

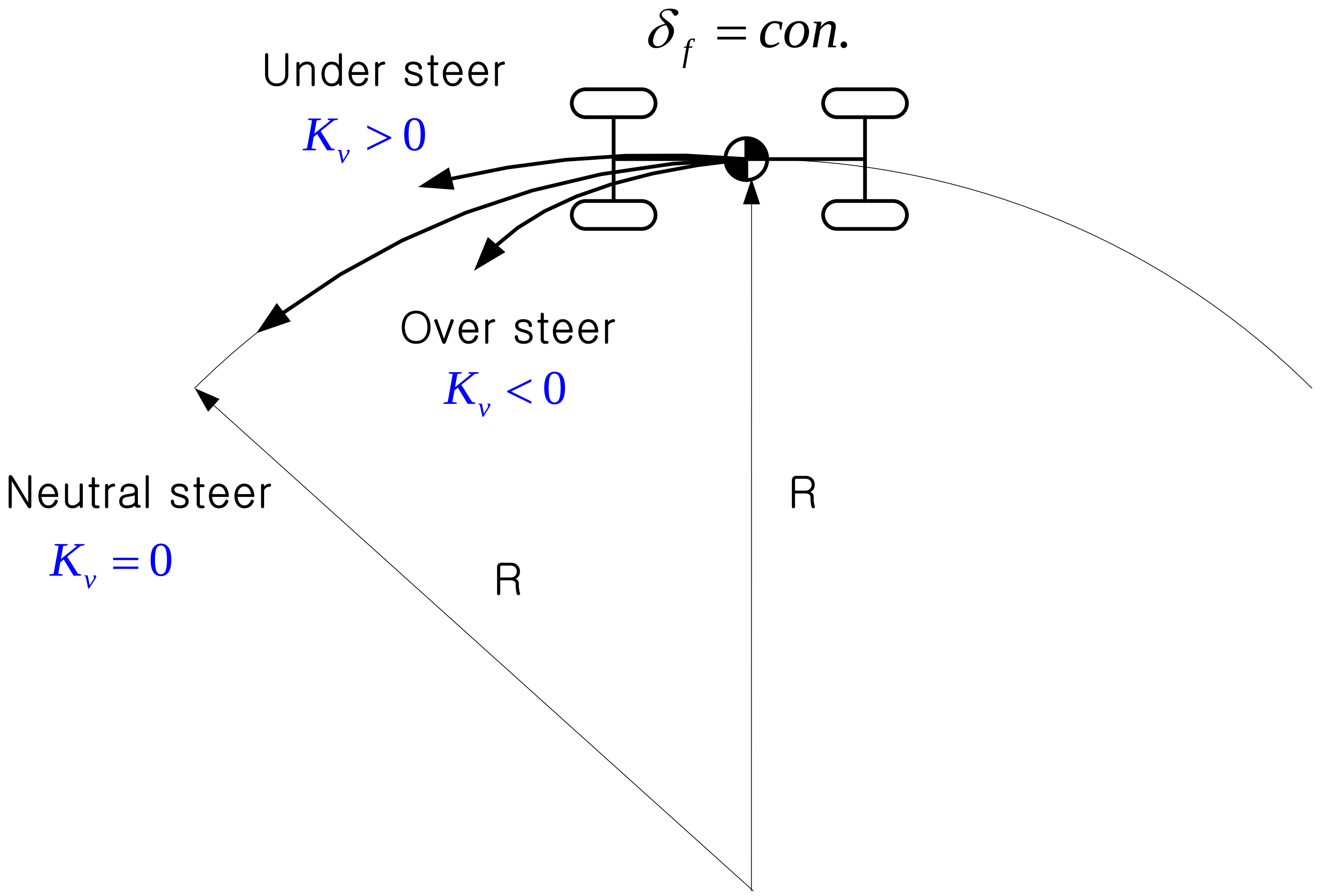
$$K_v = 0 \quad \Leftrightarrow \quad \alpha_f = \alpha_r \quad \cdots \quad \text{Neutral-steer}$$

$$K_v < 0 \quad \Leftrightarrow \quad |\alpha_f| < |\alpha_r| \quad \cdots \quad \text{Over-steer}$$

- ✓ Cars should never be designed to be over-steered at a steady state: very difficult to control
- ✓ But can still be over-steered during the transient maneuvers
- ✓ $K_v = 0$ IF the entire system is linear

Under / Neutral / Over-steer

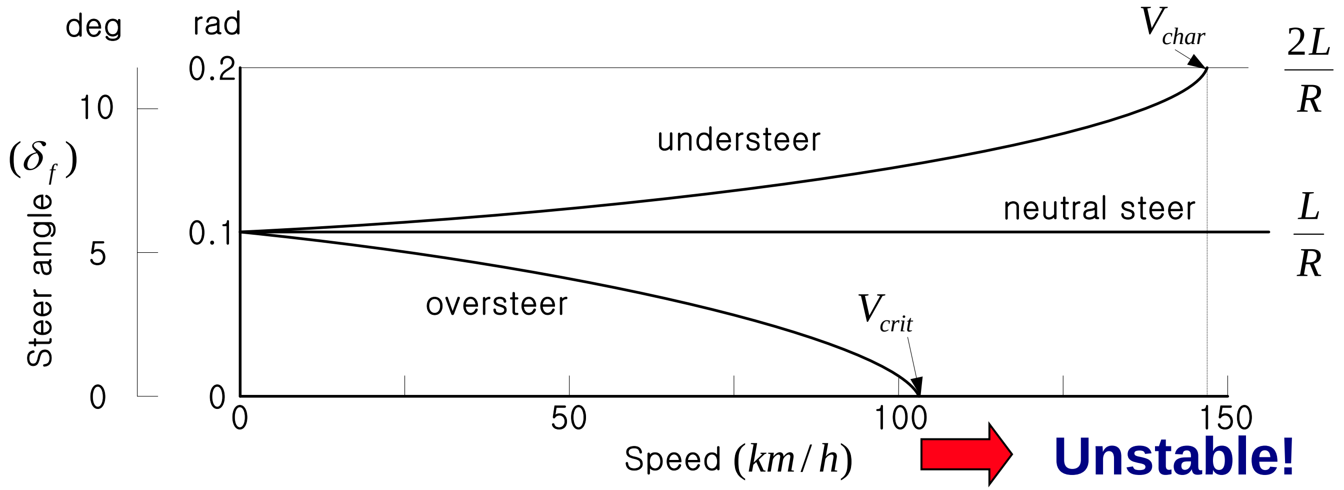
- Steady state cornering while <u>increasing speed gradually</u>



Characteristic Speed

- The vehicle speed when the steering angle is twice of Ackerman angle to make a constant radius circle
- Another expression of under-steer gradient

Constant radius R=30m, wheel base L=3m



Unstable!

✓ <u>Characteristic</u> speed (understeer) : $v_{char} = \sqrt{L/K_v}$

✓ <u>Critical</u> speed (oversteer) : $v_{crit} = \sqrt{L/(-K_v)}$

Lateral Dynamics

- Characteristic speed & steady state yaw rate

$$v_{char}^{\ 2} = \frac{L}{K_v}$$

$$\Rightarrow$$

$$\delta_f = \frac{L}{R} + K_v \frac{v_x^2}{R} = \frac{L}{R} + \frac{L}{v_{char}^2}\frac{v_x^2}{R}$$

$$= \frac{L}{R}\left[1 + \left(\frac{v_x}{v_{char}}\right)^2\right]$$

$$\Rightarrow$$

$$r = \frac{v_x}{R} = \frac{v_x}{1 + (v_x/v_{char})^2}\frac{\delta_f}{L}$$

✓ **Desired yaw rate**

Lateral Dynamics

- Over/under steering tuning: steady state response
- For steady state cornering (yaw moment balancing)

$$F_{yf} = m\frac{l_r}{L}\frac{v_x^2}{R}, \quad F_{yr} = m\frac{l_f}{L}\frac{v_x^2}{R}$$

- If $C_{f/r}$ are also proportional to normal load,

$$\Rightarrow \quad C_{f/r} = C_0 F_{zf/r}$$

$$F_{yf} = -C_f \alpha_f = -C_0 F_{zf} \alpha_f = -C_0 mg\frac{l_r}{L}\alpha_f$$

$$\Rightarrow \quad F_{yr} = -C_r \alpha_r = -C_0 F_{zr} \alpha_r = -C_0 mg\frac{l_f}{L}\alpha_r$$

$$\frac{F_{yf}}{F_{yr}} = \frac{l_r}{l_f} = \frac{l_r\alpha_f}{l_f\alpha_r} \quad \Rightarrow \quad \alpha_f = \alpha_r$$

✓ Always neutral steer for any C.G. location - if everything is linear

Lateral Dynamics

- Over/under steering tuning: steady state response
- For steady state cornering (yaw moment balancing)

$$\Rightarrow \quad F_{yf} = m\frac{l_r}{L}\frac{v_x^2}{R}, \quad F_{yr} = m\frac{l_f}{L}\frac{v_x^2}{R}$$

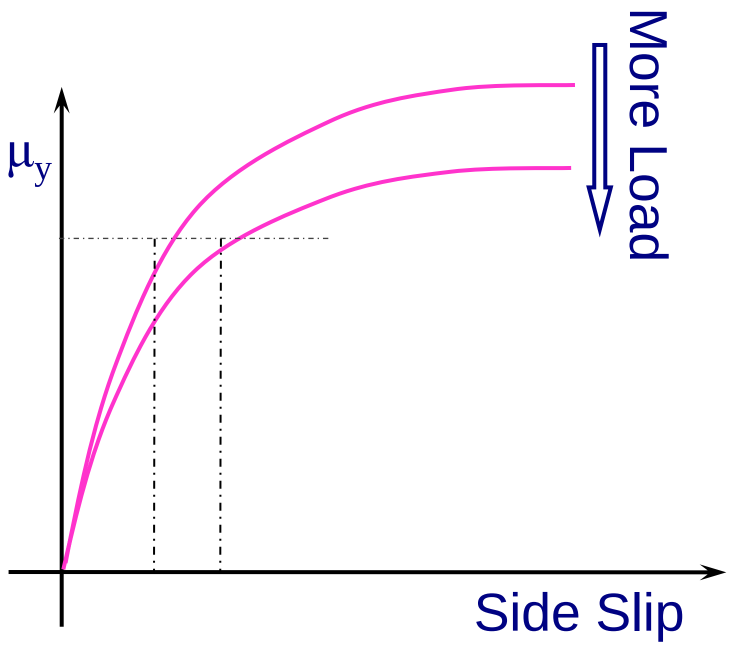
$$\frac{F_{yf}}{F_{yr}} = \frac{l_r}{l_f}$$

- For actual nonlinear tire model

$$\Rightarrow \quad F_{yf} = \mu_{yf}F_{zf}, \quad F_{yr} = \mu_{yr}F_{zr}$$

$$\frac{F_{yf}}{F_{yr}} = \frac{\mu_{yf}F_{zf}}{\mu_{yr}F_{zr}} = \frac{\mu_{yf}}{\mu_{yr}}\frac{l_r}{l_f} = \frac{l_r}{l_f}$$

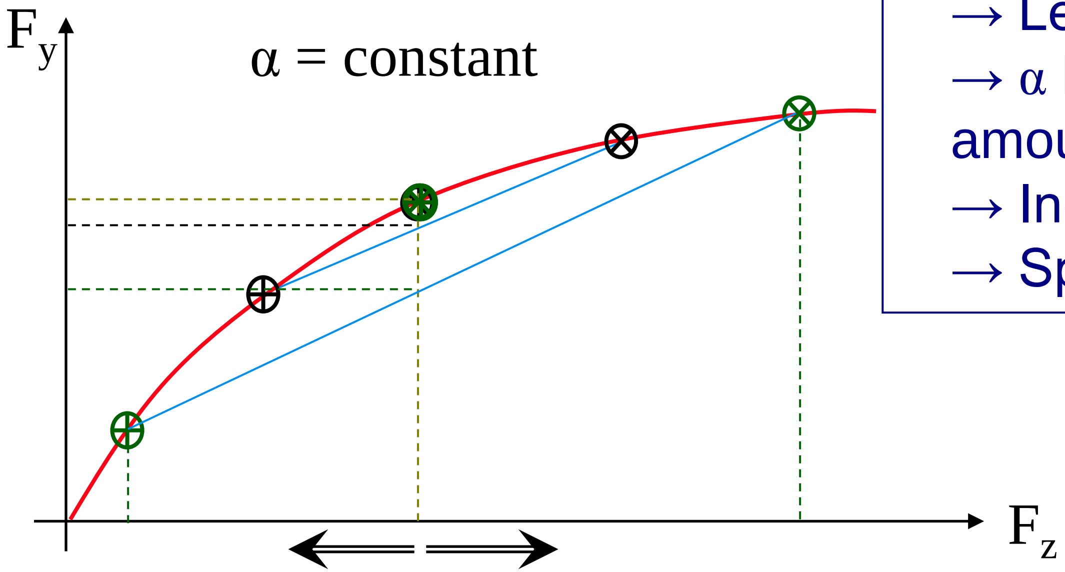
$$\Rightarrow \quad \therefore \mu_{yf} = \mu_{yr}$$



If $l_f < l_r \quad \Rightarrow \quad F_{zf} > F_{zr} \quad \Rightarrow \quad \alpha_f > \alpha_r \quad \leftarrow$ Understeer

Lateral Dynamics

- Over/under steering tuning: steady state response
  - Change CG location: hard to change especially when the vehicle design is completed
  - Front and rear different tires: Porsche
  - Tuning roll bar stiffness: Front and rear ends have same amount of roll angle, weight shifting = func. of roll stiffness



✓ Stiffer rear roll bar
→ More weight shifting at rear axle
→ Less F_y for a given α
→ α has to increase to get the right amount of F_y back
→ Increase over-steering tendency
→ Sports car

✓ Body stiffness matters

Lateral Dynamics

- Over/under steering tuning: Transient response
  - At the beginning of the turning maneuver weight is shifted to the front axle, why?
    - Vehicle C.G. does not change
    - Vehicle load allocated on each axle changes

$$F_{zf} > mg\frac{l_r}{L}, \quad F_{zr} < mg\frac{l_f}{L}$$

  - Lateral forces to balance the yaw moment is not changed(Steady-state)

$$F_{yf} = m\frac{l_r}{L}\frac{v_x^2}{R}, \quad F_{yr} = m\frac{l_f}{L}\frac{v_x^2}{R}$$

  - Apply linear cornering stiffness tire model

$$F_{yf} = -C_0 F_{zf}\alpha_f, \quad F_{yr} = -C_0 F_{zr}\alpha_r$$

$$\Rightarrow \quad |\alpha_r| > |\alpha_f| \quad \leftarrow \text{oversteer}$$

✓ Front heavy FWD cars can be oversteered during transient maneuvers